

DERIVING PROGRAM LOGICS FROM DISTRIBUTIVE MONOIDAL CATEGORIES

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MOTIVATION AND CONTEXT

categorical semantics of programs

↳ reason about program behaviour and properties
abstraction and compositionality

morphisms $\leftrightarrow$ programs

This work:

- categorical semantics of imperative programs with effects
- Inference rules derive from the categorical structure

PROGRAM SYNTAX

$C ::= \text{skip} \mid \text{abort} \mid C; C \mid \text{if } b \text{ then } C \text{ else } C \mid \text{while } b \text{ do } C$
 $\mid \text{assert } p \mid x_i \leftarrow e \mid (x_{i_1}, \dots, x_{i_k}) \leftarrow s$

guards may be nondeterministic/probabilistic

IMPERATIVE CATEGORIES

Denotational semantics of effectful imperative programs.

Imperative category = distributive copy-discard category with a compatible uniform coproduct-trace.

- coproducts
↳ control-flow

$$^{(+)} \quad -b \left\{ \begin{array}{l} f \\ g \end{array} \right. = \text{if } b \text{ then } f \text{ else } g$$

- copy-discard monoidal
↳ data-flow
variable sharing

$$^{(\otimes)} \quad \begin{array}{c} \text{---} f \text{---} \\ \text{---} g \text{---} \end{array} = \begin{array}{l} x_1 \leftarrow f(x_1, x_2); \\ (x_2, x_3) \leftarrow g(x_1, x_2) \end{array}$$

- uniform coproduct-trace
↳ while loops

$$^{(+)} \quad \begin{array}{c} \text{---} b \text{---} f \text{---} \\ \text{---} \end{array} = \text{while } b \text{ do } f$$

[cf. Arbib, Manes (1980), Lockett (1993), Jhasegawa (1997), Jacobs (2015), ...]

EXAMPLES

- Par , deterministic semantics
↳ sets and partial functions
- Rel , relational semantics
↳ sets and relations
- $\text{Stoch}_{\leq 1}$, discrete probabilistic semantics
↳ sets and countably supported conditional subdistribution
- $\text{BorelStoch}_{\leq 1}$, measurable probabilistic semantics
↳ standard Borel spaces and measurable conditional subdistributions

DERIVING PROGRAM LOGICS

$\{P\} c \{Q\}$

precondition $\{P\}$ c $\{Q\}$ postcondition

program

- Correctness interpretation
"If the inputs satisfy P , then the outputs satisfy Q "
- Incorrectness interpretation
"If the outputs satisfy Q , then the inputs satisfy P "

Define the validity of triples w.r.t. their semantics in poset-enriched imperative categories:

	$P, Q : 1 \rightarrow X$	$P, Q : X \rightarrow 1$
CORRECTNESS	$\llbracket P \rrbracket; \llbracket c \rrbracket \leq \llbracket Q \rrbracket$	$\llbracket P \rrbracket \leq \llbracket c \rrbracket; \llbracket Q \rrbracket$
INCORRECTNESS	$\llbracket P \rrbracket; \llbracket c \rrbracket \geq \llbracket Q \rrbracket$	$\llbracket P \rrbracket \geq \llbracket c \rrbracket; \llbracket Q \rrbracket$

EXAMPLES

CORRECTNESS

$P, Q : 1 \rightarrow X$

INCORRECTNESS

$P, Q : 1 \rightarrow X$

CORRECTNESS

$P, Q : X \rightarrow 1$

INCORRECTNESS

$P, Q : X \rightarrow 1$

Rel

Hoare triples

[Hoare (1969)]

incorrectness triples

[de Vries, Koutavas (2011)]

weakest preconditions

strongest preconditions

Stoch_{≤1}

[...]

[...]

weakest preexpectations

[B.K.K.M. (2021)]

strongest preexpectations

CONCLUSIONS

- We propose imperative categories as denotational semantics of imperative programs with effects
- We derive the rules of multiple program logics from the categorical structure

MORE IN THE PAPER

- A monoidal guarded command language and its algebra (cf. GKAT [Smolka, Foster, Jhu, Kaprie, Kozen, Silva (2019)])
- Relational program logics

FUTURE WORK

- More expressive assertions: (nonconvex) sets of distributions
- Relative completeness

EXAMPLE : LOOP INVARIANTS

$$\frac{[\text{assert } p]; [c] \leq [c]; [\text{assert } p]}{[\text{assert } p]; [\text{while } b \text{ do } c] \leq [\text{while } b \text{ do } c]; [\text{assert } p]}$$

PROOF

$$\begin{aligned}
 & \text{(coproducts)} \quad \text{(distributivity)} \quad \text{(Hp)} \\
 & \frac{-p}{-p} \} b \frac{c}{-} = \frac{-p}{-p} \} p - b \frac{c}{-} = \frac{-p}{-p} \} b \frac{p-c}{p-} \leq \frac{-p}{-p} \} b \frac{c-p}{p-} \\
 & \text{(uniformity)} \Rightarrow \frac{-p}{-p} \} b \frac{c}{-} \leq \frac{-p}{-p} \} b \frac{c}{p-}
 \end{aligned}$$

□

EXAMPLE : GUARDED DISJUNCTION

$$\frac{[s_1]; [c] \leq [t_1] \quad [s_2]; [c] \leq [t_2]}{[s_1 +_b s_2]; [c] \leq [t_1 +_b t_2]}$$

PROOF

$$\begin{array}{ccccc} & \text{(coproducts)} & & \text{(Hp)} & \\ & \downarrow & & \downarrow & \\ -b \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} - c - & = & -b \begin{bmatrix} s_1 - c \\ s_2 - c \end{bmatrix} - & \leq & -b \begin{bmatrix} t_1 \\ t_2 \end{bmatrix} - \end{array}$$

□